Prognostic Factors in Melanoma

Microstaging (See Table 124-4)

TUMOR THICKNESS

The single most important prognostic factor for survival and clinical management in localized stage I and II cutaneous melanoma is **tumor thickness**.[68] As originally described by Breslow, tumor thickness is measured from the top of the granular layer of the epidermis to the greatest depth of tumor invasion, using an ocular micrometer and reported in millimeters (see eFig. 124-17.1 in online edition). Survival decreases with increasing Breslow depth.

Clark level is an alternate, less accurate method of assessing tumor invasion based on anatomic level and is no longer the primary criterion in modern staging. Historically described by Clark, the levels are: - **Level I**: Melanoma confined to the epidermis (in situ) - **Level II**: Extension into the papillary dermis - **Level III**: Tumor fills and expands the papillary dermis - **Level IV**: Invasion into the reticular dermis - **Level V**: Invasion into the subcutaneous fat

While Clark level may influence prognosis in thin melanomas (<1 mm), where levels IV or V can upstage a tumor from T1a to T1b, evidence suggests it is a weaker prognostic indicator compared to Breslow thickness.[69]

ULCERATION

Ulceration is currently recognized as an independent prognostic factor in primary melanoma. Its presence worsens prognosis and is incorporated into the AJCC staging system. Ulceration is defined histologically as disruption of the epidermis overlying the melanoma, associated with tumor-induced necrosis rather than trauma. It carries significant weight in microstaging and upstaging of melanomas, particularly in tumors of intermediate thickness.